

CS4234: Optimization Algorithms
Semester 1, AY 2026-27
School of Computing, NUS

Solutions for Tutorial 2

Question 1: Traveling Salesperson Algorithms

1A. An $O(2^n n^2)$ dynamic program

Fix a vertex $r \in V$. For every subset $S \subseteq V$ with $r \in S$ and every $v \in S$, define

$D[S, v]$ = minimum cost of a path that starts at r , visits exactly the vertices in S , and ends at v .

The path is required to visit every vertex in S exactly once. The base case is

$$D[\{r\}, r] = 0.$$

All other states that do not satisfy the definition are assigned value $+\infty$. For $|S| \geq 2$ and $v \in S \setminus \{r\}$, the recurrence is

$$D[S, v] = \min_{u \in S \setminus \{v\}} (D[S \setminus \{v\}, u] + c(u, v)).$$

After filling the table in increasing order of $|S|$, the optimal tour cost is

$$\min_{v \in V \setminus \{r\}} (D[V, v] + c(v, r)).$$

Correctness. We prove the recurrence by induction on $|S|$. The base state is the empty path at r . Consider an optimal path counted by $D[S, v]$, and let u be the vertex immediately before v . Removing its last edge (u, v) leaves a minimum-cost path counted by $D[S \setminus \{v\}, u]$; otherwise replacing that prefix by a cheaper one would improve the original path. Conversely, appending (u, v) to any path counted by $D[S \setminus \{v\}, u]$ produces a feasible path for $D[S, v]$. Taking the minimum over all possible predecessors u is therefore exact. Finally, every Hamiltonian cycle through r has some last vertex v before returning to r , which gives the final minimization above.

There are at most $n2^{n-1} = O(n2^n)$ states and at most n transitions per state. Thus the running time is $O(2^n n^2)$ and the table uses $O(2^n n)$ space. Storing a minimizing predecessor for each state also allows the tour to be reconstructed.

1B. A metric instance where double-tree is not optimal

Consider four vertices arranged on the unit cycle

$$A - B - D - C - A.$$

Give each cycle edge cost 1, and give the two diagonals $A - D$ and $B - C$ cost 2. These costs form a metric: they are exactly the shortest-path distances in the unit cycle.

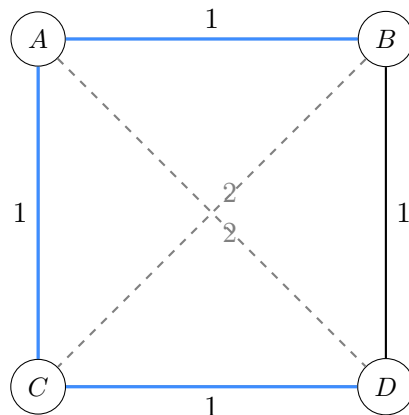


Figure 1: A metric four-vertex instance. One possible MST is highlighted in blue.

Choose the MST with edges $A - B$, $A - C$, and $C - D$, which has cost 3. After doubling these edges, one valid Eulerian tour is

$$A, B, A, C, D, C, A.$$

Shortcutting repeated vertices returns

$$A, B, C, D, A,$$

whose cost is

$$1 + 2 + 1 + 2 = 6.$$

However,

$$A, B, D, C, A$$

is a valid tour of cost

$$1 + 1 + 1 + 1 = 4.$$

Thus this valid execution of the double-tree algorithm is not optimal.

Question 2: Building Roads for Eulerian Tours

Use the abbreviations M (Motel Mildred), N (Nice View), C (Famous Coffee), P (Central Plaza), T (Tram Station), H (Hotel Humperdink), and S (South Junction). Their initial degrees are

vertex	M	N	C	P	T	H	S
deg	2	2	4	5	4	2	3

Thus the two initially odd vertices are P and S . Adding a road toggles the parity of exactly its two endpoints.

2A. Humperdink's road

To obtain an Eulerian path from H to P , the odd vertices must become precisely H and P . Vertex P is already odd, so its parity should not change. We must toggle H from even to odd and S from odd to even. Therefore, Humperdink should build a **second, parallel road between H and S** .

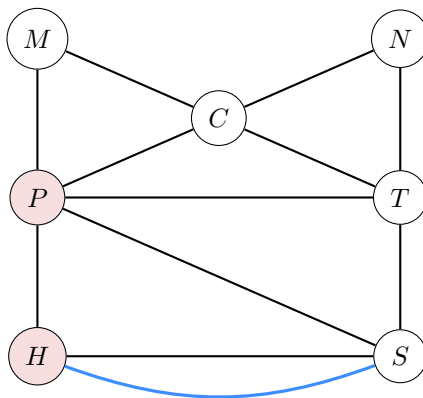


Figure 2: Part 2A: the blue curved edge is a new road parallel to the existing H – S road. The remaining odd vertices are H and P .

After this addition, $\deg(H) = 3$ and $\deg(S) = 4$, while all other degrees are unchanged. Hence exactly H and P are odd, as required.

For example, treating the two parallel H – S roads as distinct edges, one valid Eulerian path is

$$H, P, M, C, N, T, C, P, T, S, H, S, P.$$

2B. Mildred's road

After Part 2A, the odd vertices are H and P . Mildred wants them to be M and P . She should therefore toggle H and M by building a **road between Motel Mildred and Hotel Humperdink**.

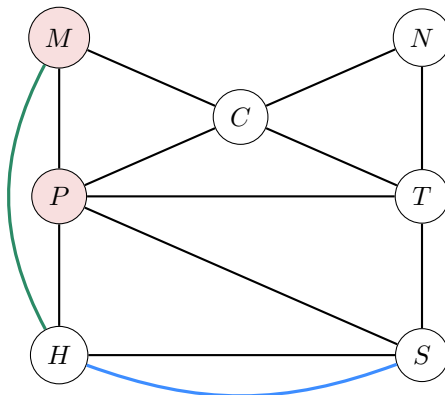


Figure 3: Part 2B: the green road M – H moves the odd endpoint from H to M . The blue road from Part 2A remains.

Now M and P are the only odd vertices, so an Eulerian path runs from M to P . Vertex H is even, so no Eulerian path can start at H and end at P .

One valid Eulerian path after both roads have been added is

$$M, H, P, M, C, N, T, C, P, T, S, H, S, P.$$

2C. Characterization of Eulerian paths

Necessity. Suppose an Eulerian path starts at s and ends at d . At every vertex other than s and d , each arrival along an edge is paired with a departure along a different edge. Thus its incident edges occur in pairs, so the vertex has even degree. At s , there is one initial departure with no matching arrival; all other incident edges are paired. Hence s has odd degree. Similarly, d has one final arrival with no matching departure, so d has odd degree.

Sufficiency. Suppose s and d are odd and every other vertex is even. Add one new edge between s and d . The resulting connected multigraph has even degree at every vertex, so Euler's cycle theorem (or Hierholzer's algorithm) gives an Eulerian cycle. Delete the newly added edge from that cycle and start reading the remaining edges immediately after the deletion. What remains is a trail that starts at one of s, d , ends at the other, and uses every original edge exactly once. Reverse it if necessary to obtain an Eulerian path from s to d .

Question 3: Group Admission at a Museum

Groups larger than n cannot occur in any feasible solution, so they may be ignored in every part.

3A. The largest-first algorithm is a $1/2$ -approximation

Let A be the number of visitors admitted by the greedy algorithm.

If $A \geq n/2$, then $\text{OPT} \leq n$ immediately gives

$$A \geq \frac{n}{2} \geq \frac{\text{OPT}}{2}.$$

It remains to consider $A < n/2$. The final unused capacity is then

$$r = n - A > \frac{n}{2}.$$

Remaining capacity never increases, so it was at least r whenever a group was processed. Therefore every group of size at most $n/2$ fitted when it was considered and was admitted.

There also cannot be a feasible group of size strictly greater than $n/2$. If such groups existed, the largest-first order would consider one of them before any group of size at most $n/2$. The first feasible such group would fit into the initially empty museum, forcing $A > n/2$, a contradiction. Thus every group that could belong to an optimal solution was admitted by the greedy algorithm. In this case $A = \text{OPT}$.

In both cases, $A \geq \text{OPT}/2$.

3B. A family approaching the factor $1/2$

Let $n = 2k$ and consider three groups of sizes

$$k + 1, \quad k, \quad k.$$

The greedy algorithm first admits the group of size $k + 1$. Only $k - 1$ places remain, so it rejects both groups of size k . Hence

$$\text{ALG} = k + 1.$$

The optimum admits the two groups of size k , filling all $2k$ places, so

$$\text{OPT} = 2k.$$

Consequently,

$$\frac{\text{ALG}}{\text{OPT}} = \frac{k + 1}{2k} = \frac{1}{2} + \frac{1}{2k} \longrightarrow \frac{1}{2}.$$

3C. An $O(m)$ -time, $O(1)$ -space algorithm

Make one pass through the read-only array to find an index b of a largest group satisfying $g_b \leq n$. If no such group exists, return the empty solution.

1. If $g_b \geq n/2$, admit only group b .
2. If $g_b < n/2$, make a second pass through the groups in their given order, admitting a group whenever it fits in the remaining capacity.

The algorithm stores only the current maximum, its index, the remaining capacity, and the running admitted total, so it uses $O(1)$ extra space.

For the first case,

$$\text{ALG} = g_b \geq \frac{n}{2} \geq \frac{\text{OPT}}{2}.$$

For the second case, every feasible group has size strictly less than $n/2$. Let A be the total produced by the second pass. If $A \geq n/2$, the same capacity bound proves $A \geq \text{OPT}/2$. If $A < n/2$, the final unused capacity is greater than $n/2$ and was never smaller earlier in the scan. Every feasible group is smaller than this amount, so every feasible group was admitted. The solution is then optimal. Thus the algorithm is always a $1/2$ -approximation.

Both passes take $O(m)$ time, and the extra space is $O(1)$.